

BioRED: a rich biomedical relation extraction dataset

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Abstract

Automated relation extraction (RE) from biomedical literature is critical for many downstream text mining applications in both research and real-world settings. However, most existing benchmarking datasets for biomedical RE only focus on relations of a single type (e.g. protein–protein interactions) at the sentence level, greatly limiting the development of RE systems in biomedicine. In this work, we first review commonly used named entity recognition (NER) and RE datasets. Then, we present a first-of-its-kind biomedical relation extraction dataset (BioRED) with multiple entity types (e.g. gene/protein, disease, chemical) and relation pairs (e.g. gene–disease; chemical–chemical) at the document level, on a set of 600 PubMed abstracts. Furthermore, we label each relation as describing either a novel finding or previously known background knowledge, enabling automated algorithms to differentiate between novel and background information. We assess the utility of BioRED by benchmarking several existing state-of-the-art methods, including Bidirectional Encoder Representations from Transformers (BERT)-based models, on the NER and RE tasks. Our results show that while existing approaches can reach high performance on the NER task (F-score of 89.3%), there is much room for improvement for the RE task, especially when extracting novel relations (F-score of 47.7%). Our experiments also demonstrate that such a rich dataset can successfully facilitate the development of more accurate, efficient and robust RE systems for biomedicine.

Availability: The BioRED dataset and annotation guidelines are freely available at <https://ftp.ncbi.nlm.nih.gov/pub/lu/BioRED/>.

Keywords: relation extraction, named entity recognition, biomedical dataset, biomedical natural language processing

Introduction

Biomedical natural language processing (BioNLP) and text-mining methods/tools make it possible to automatically unlock essential information published in the medical literature, including genetic diseases and their relevant variants [1, 2], chemical-induced diseases [3] and drug response in cancer [4]. However, two crucial and building block steps in the general biomedical information extraction pipeline remain challenging. The first is named entity recognition and linking (NER/NEL), which automatically recognizes the boundary of the entity spans (e.g. ESR1) of a specific biomedical concept (e.g. gene) from the free text and further links the spans to the particular entities with database identifiers (e.g. NCBI Gene ID: 2099). The second is relation extraction (RE), which identifies an entity pair with certain relations.

To facilitate the development and evaluation of NLP and machine learning methods for biomedical NER/NEL and RE, significant efforts have been made on relevant corpora development [5–10]. However, most existing corpora focus only on relations between two entities and within single sentences. For example, Herrero-Zazo *et al.*

[8] developed a drug–drug interaction (DDI) corpus by annotating relations only if both drug names appear in the same single sentence. As a result, multiple individual NER/RE tools need to be created to extract biomedical relations beyond a single type (e.g. extracting both DDI and gene–disease relations).

Additionally, in the biomedical domain, extracting novel findings that represent the fundamental reason why an asserted relation is published as opposed to background or ancillary assertions from the scientific literature is of significant importance. To the best of our knowledge, none of the previous works on (biomedical) relation annotation, however, included such a novelty attribute.

In this work, we first give an overview of NER/NEL/RE datasets and show their strengths and weaknesses. Furthermore, we present a rich biomedical relation extraction dataset (BioRED). We further annotated the relations as either novel findings or previously known background knowledge. We summarize the unique features of the BioRED corpus as follows: (i) BioRED consists of biomedical relations among six commonly

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Mutations in the PCSK9 gene in Norwegian subjects with autosomal dominant hypercholesterolemia.

abstract offset: 96 - 1112

Proprotein convertase subtilisin/kexin type 9 (PCSK9) is at a locus for autosomal dominant hypercholesterolemia, and recent data indicate that the PCSK9 gene is involved in cholesterol biosynthesis. Mutations within this gene have previously been found to segregate with hypercholesterolemia. In this study, DNA sequencing of the 12 exons of the PCSK9 gene has been performed in 51 Norwegian subjects with a clinical diagnosis of familial hypercholesterolemia where mutations in the low-density lipoprotein receptor gene and mutation R3500Q in the apolipoprotein B-100 gene had been excluded. Two novel missense mutations were detected in the catalytic subdomain of the PCSK9 gene. Two patients were heterozygotes for D374Y, and one patient was a double heterozygote for D374Y and N157K. D374Y segregated with hypercholesterolemia in the two former families where family members were available for study. Our findings support the notion that mutations in the PCSK9 gene cause autosomal dominant hypercholesterolemia.

Relation #R5

Note: Novel

Relation Type: Positive_Correlation

Type	Concept ID	Text
Gene	255738	PCSK9
Disease	D006938	autosomal dominant hypercholesterolemia
Chemical	D002784	cholesterol
Disease	D006937	hypercholesterolemia
Gene	3949	low-density lipoprotein receptor
Variant	rs5742904	R3500Q
Gene	338	apolipoprotein B-100
Species	9606	patients
Variant	rs137852912	D374Y
Variant	rs143117125	N157K

Figure 1. An example of a relation and the relevant entities displayed on TeamTat (<https://www.teamtat.org>).

described entities (i.e. gene, disease, chemical, variant, species and cell line) in eight different types (e.g. positive correlation). Such a setting supports developing a single general-purpose RE system in biomedicine with reduced resources and improved efficiency. More importantly, several previous studies have shown that training a machine learning algorithm on multiple concepts simultaneously on one dataset, rather than multiple single-entity datasets, can lead to better performance [11–13]. We expect similar outcomes with our dataset for both NER and RE tasks. (ii) The annotated relations can be asserted either within or across sentence boundaries. For example, as shown in Figure 1 (relation R5 in pink), the variant ‘D374Y’ of the PCSK9 gene and the causal relation with the disease ‘autosomal dominant hypercholesterolemia’ are in different sentences. This task, therefore, requires relations to be inferred by machine reading across the entire document. (iii) Finally, our corpus is enriched with novelty annotations. This novel task poses new challenges for (biomedical) RE research and enables the development of NLP systems to distinguish between known facts and novel findings, a greatly needed feature for extracting new knowledge and avoiding duplicate information toward the automatic knowledge construction in biomedicine.

To assess the challenges of BioRED, we performed benchmarking experiments with several state-of-the-art methods, including Bidirectional Encoder Representations from Transformers (BERT)-based models. We find that existing deep-learning systems perform well on the NER task but only modestly on the novel RE task, leaving it an open problem for future NLP research. Furthermore, the detailed analysis of the results confirms the benefit of using such a rich dataset toward creating more accurate, efficient and robust RE systems in biomedicine.

Overviews of NER/NEL/RE datasets

Named entity recognition and named entity linking

Existing NER/NEL datasets cover most of the key biomedical entities, including gene/proteins [14–16], chemicals [17, 18], diseases [9, 19], variants [20–22], species [23, 24] and cell lines [25]. Nonetheless, NER/NEL datasets usually focus on only one concept type; the very few datasets that annotate multiple concept types [26, 27] do not contain relation annotations. Table 1 summarizes some widely used gold standard NER/NEL datasets, including the annotation entity type, corpus size and the task applications.

Due to the limitation of the entity type in NER datasets, most of the state-of-the-art entity taggers were developed individually for a specific concept. A few studies (e.g. PubTator [28]) integrate multiple entity taggers and apply them to specific collections or even to the entire PubMed/PubMed Central (PMC). In the development process, some challenging issues related to integrating entities from multiple taggers, such as concept ambiguity and variation emerged [29]. Moreover, the same articles need to be processed multiple times by multiple taggers. Huge storage space also is required to store the results of the taggers. In addition, based on clues from previous NER studies [12, 30], we realized that a tagger trained with other concepts performs as well or even better than a tagger trained on only a single concept, especially for highly ambiguous concepts. A gene tagger GNormPlus trained on multiple relevant concepts (gene/family/domain) boosts the performance of a gene/protein significantly. Therefore, a rich NER corpus can help develop a method that can recognize multiple entities simultaneously to reduce the hardware requirement and achieve better performance. Only a very few datasets [5,

Table 1. Overview of gold standard NER/NEL datasets

Dataset	Text size	Entity type (#mentions)	Task type
JBLPBA [26]	2404 abstracts	Protein (35 336), DNA (10 589), RNA (1069), cell line (4330) and cell type (8639)	NER
NCBI Disease [19]	793 abstracts	Disease (6892)	NER, NEL
CHEMDNER [18]	10 000 abstracts	Chemical (84 355)	NER
BC5CDR [9]	1500 abstracts	Chemical (15 935), Disease (12 850)	NER, NEL
LINNAEUS [24]	100 PMC full text	Species (4259)	NER, NEL
tmVar [20]	500 abstracts	Variant (1431)	NER, NEL
NLM-Gene [14]	550 abstracts	Gene (15 553)	NER, NEL
GNormPlus [12]	694 abstracts	Gene (9986)	NER, NEL

27] curate multiple concepts in the text, but no relation is curated in these datasets.

Relation extraction

A variety of RE datasets in the general domain have been constructed to promote the development of RE systems [31–33]. Many of the RE datasets focus on extracting relations from a single sentence. Since many relations cross sentence boundaries, moving research from the sentence level to the document level (e.g. DocRED [34], DocOIE [35]) became a popular trend recently. In the biomedical domain, most existing RE datasets [6, 8, 10] focus on sentence-level relations involving a single pair of entities. However, multiple sentences are often required to describe an entire biological process or a relation. We highlight several commonly used biomedical RE datasets in Table 2 (a complete dataset review can be found in Table S1, see Supplementary Data available online at <http://bib.oxfordjournals.org/>). But only very few datasets contain relations across multiple sentences (e.g. BC5CDR dataset [9]). Most of the datasets [6–10, 36–40], which were widely used for the RE system development [41–45], focus on the single entity pair only (e.g. AIMed [37] to protein–protein interaction). Some of those datasets annotated the relation categories more granular. For example, DDI13 [8] annotated 4 categories (i.e. advise, int, effect and mechanism) of the DDI, ChemProt [10] annotated 5 categories of the chemical–protein interaction and DrugProt [40], an extension of ChemProt, annotated 13 categories. Recently, ChemProt and DDI13 are widely used in evaluating the abilities of biomedical pretrained language models [46–49] on RE tasks.

During the curation of the relations at the sentence level, curators usually do not access the context of the surrounding sentences. Besides, most sentence-level RE datasets do not link the entity names to the concept identifiers (e.g. NCBI Gene ID) in the external resources/databases. Instead, the RE dataset development at the document level relies highly on the concept identifiers. But it is extremely time-consuming, and very limited biomedical datasets annotate the related entities to the concept identifiers. BC5CDR dataset [9] is a widely used dataset with chemical-induced disease relations at the document level. All of the chemicals and diseases are linked to the concept identifiers. However, BC5CDR did

not annotate the relations (e.g. treatment) out of the chemical-induced disease category. Peng et al. [56] developed a cross-sentence n-ary relation extraction dataset with the relations among drug, gene and mutation. But the dataset is constructed via distant supervision with the inevitable wrong labeling problem [34] instead of manual curation. Moreover, BioNLP shared task datasets [59–62] provide fine-grained biological event annotations to promote biological activity extraction. In Table 3, we compare BioRED to representative biomedical relation extraction datasets. BioRED covers more types of entity pairs than those datasets.

Methods

Annotation definition/scope

We first analyzed a set of public PubMed search queries by tagging different entities and relations. This data-driven approach allowed us to determine a set of key entities and relations of interest that should be most representative, and therefore the focus of this work. Some entities are closely related biologically and are thus used interchangeably in this work. For instance, protein, mRNA and some other gene products typically share the same names and symbols. Thus, we merged them to a single gene class, and similarly merged symptoms and syndromes to a single disease class. In the end, we have six concept types: (i) Gene: for genes, proteins, mRNA and other gene products; (ii) Chemical: for chemicals and drugs; (iii) Disease: for diseases, symptoms and some disease-related phenotypes; (iv) Variant: for genomic/protein variants (including substitutions, deletions, insertions and others); (v) Species: for species in the hierarchical taxonomy of organisms and (vi) CellLine: for cell lines. Due to the critical problems of term variation and ambiguity, entity linking (also called entity normalization) is also required. We linked the entity spans to specific identifiers in an appropriate database or controlled vocabulary for each entity type (e.g. NCBI Gene ID for genes).

Between any of two different entity types, we further observed eight popular associations that are frequently discussed in the literature: <D,G> for <Disease, Gene>; <D,C> for <Disease, Chemical>; <G,C> for <Gene, Chemical>; <G,G> for <Gene, Gene>; <D,V> for

Table 2. A summary of biomedical RE and event extraction datasets. The value of ‘-’ means that we could not find the number in their papers or websites. The SEN/DOC Level means whether the relation annotation is annotated in ‘Sentence’, ‘Document’ or ‘Cross-sentence’. ‘Document’ includes abstract, full-text or discharge record. ‘Cross-sentence’ allows two entities within a relation to appear in three surrounding sentences

Datasets	# Doc./Sent.	# Entities	# Relations	SEN/DOC Levels	Descriptions
Protein–protein interaction AImed [37]	230 abstracts	4141 genes	1101 relations	Sentence	The AImed dataset aims to develop and evaluate protein name recognition and protein–protein interaction (PPI) extraction. It contains 750 Medline abstracts, which contain the ‘human’ word, and has 5206 names. Two hundred abstracts previously known to contain protein interactions for PPI extraction were obtained from the Database of Interacting Proteins (DIP) [50] and tagged for both 1101 protein interactions and 4141 protein names. Because negative examples for protein interactions were rare in the 200 abstracts, they manually selected 30 additional abstracts with more than one gene but did not have any gene interactions.
BioInfer [6]	1100 sentences	4573 proteins	2662 relations	Sentence	A PPI dataset uses ontologies defining the fine-grained types of entities (like ‘protein family or group’ and ‘protein complex’) and their relationships (like ‘CONTAIN’ and ‘CAUSE’). They developed a corpus of 1100 sentences containing full dependency annotation, dependency types and comprehensive annotation of bio-entities and their relationships.
BioCreative II PPI IPS [7]	1098 full-texts	-	-	Document	The BioCreative II PPI protein interaction pairs subtask (IPS) provides 750 and 356 full texts for training and test sets, respectively. The full text includes corresponding gene mention symbols and PPI pairs.
Chemical–protein interaction DrugProt [40]	5000 abstracts	65 561 chemicals, 61 775 genes	24 526 relations	Sentence	The DrugProt dataset aims to promote the development of chemical–gene RE systems, an extension of the ChemProt dataset. It addresses 13 different chemical–gene relations, including regulatory, specific and metabolic relations
Chemical–disease interaction BC5CDR [9]	1500 abstracts	15 935 chemicals; 12 850 diseases	3106 relations	Document	BC5CDR consists of 1500 abstracts that chemical and disease mention annotations and their IDs. It annotates chemical-induced disease relation ID pair. There are 1400 abstracts selected from a CTD–Pfizer collaboration-related dataset, and the remaining 100 articles are new curation and are used in the test set.
DDI and Drug–ADE(adverse drug effect) interaction ADE [51]	2972 MEDLINE case report	5063 drugs; 5776 adverse effects; 231 dosages	6821 drug–adverse effects; 279 drug–dosage relations	Sentence	The ADE dataset contains drugs and conditions. But the entities do not link to the standard database identifiers. Like most of the relation datasets, ADE annotates the relations (i.e. drug–ADE and drug–dosage relations) at the sentence level.
DDI13 [8]	905 documents	13 107 drugs	5028 relations	Sentence	SemEval 2013 DDIE extraction dataset consists of 792 texts selected from the DrugBank database and 233 Medline abstracts. The corpus is annotated with 18 502 pharmacological substances and 5028 DDIs, including both pharmacokinetic (PK) and pharmacodynamic (PD) interactions.
n2c2 2018 ADE [52]	505 summaries	83 869 entities	59 810 relations	-	The discharge summaries are from the clinical care database of the MIMIC-III (Medical Information Mart for Intensive Care-III). The summaries are manually selected to contain at least one ADE and annotated with nine concepts and eight relation pairs. The data are split into 303 and 202 for training and test sets, respectively.
Variant/gene–disease interaction EMU [21]	110 abstracts	-	179 relations	Document	The EMU dataset focuses on finding relationships between mutations and their corresponding disease phenotypes. They use ‘MeSH = mutation’ to select abstracts and use MetaMap [53] to annotate the abstracts that are divided into containing mutations related to prostate cancer (PCa) and breast cancer (BCa). They then use rules and patterns to select subsets of PCa and BCa for annotating.
RENET2 [54]	1000 abstracts, 500 full-texts	-	-	Document	It contains both 1000 abstracts (from RENET [55]) and 500 full texts from PMC open-access subset. For better quality, 500 abstracts of the dataset were refined. The authors used the 500 abstracts to train the RENET2 model and conduct their training data expansion using the other 500 abstracts. They further used the model trained on 1000 abstracts to construct 500 full-text articles.

continued

Table 2. Continued

Datasets	# Doc./Sent.	# Entities	# Relations	SEN/DOC Levels	Descriptions
Drug-gene mutation N-ary [56]	-	-	3462 triples; 137 469 drug-gene relations; 3192 drug-mutation relations;	Cross-sentence	Authors use distant supervision to construct a cross-sentence drug-gene mutation RE dataset. They use 59 distinct drug-gene mutation triples from the knowledge bases to extract 3462 ternary positive relation triples. The negative instances are generated by randomly sampling the entity pairs/triples without interaction.
Event extraction GE09 [57]	1200 abstracts	-	13 623 events	Sentence	As the first BioNLP shared task (ST), it aimed to define a bounded, well-defined GENIA event extraction (GE) task, considering both the actual needs and the state-of-the-art in bio-TM technology and to pursue it as a community-wide effort.
GE11 [58]	1210 abstracts, 21 616 proteins 14 full-text	18 047 events		Sentence	The BioNLP ST 2011 GE task follows the task definition of the BioNLP ST 2009, which is briefly described in this section. BioNLP ST 2011 took the role of measuring the progress of the community and generalization IE technology to the full papers.
CG [59]	600 abstracts	21 683 entities	17 248 events; 917 relations	Sentence	The BioNLP ST 2013 Cancer Genetics (CG) corpus contains annotations of over 17 000 events in 600 documents. The task addresses entities and events at all levels of biological organization, from the molecular to the whole organism, and involves pathological and physiological processes.

Table 3. Comparison of the BioRED corpus with representative relation extraction datasets

	<D,G>	<D,C>	<D,V>	<C,C>	<C,G>	<G,G>	<V,C>	<V,V>
RENET2	✓							
BC5CDR		✓						
EMU			✓					
DDI13				✓				
DrugProt					✓			
AIMed						✓		
GE11						✓		
N-ary					✓		✓	
CG	✓	✓		✓	✓	✓		
BioRED	✓	✓	✓	✓	✓	✓	✓	✓

D = Disease, G = Gene, C = Chemical and V = Variant.

<Disease, Variant>; <C,V> for <Chemical, Variant>; <C,C> for <Chemical, Chemical> and <V,V> for <Variant, Variant>. For relations between more than two entities, we simplified the relation to multiple relation pairs. For example, we simplified the chemicals co-treat disease relation ('bortezomib and dexamethasone co-treat multiple myeloma') to three relations: <bortezomib, multiple myeloma, treatment>, <dexamethasone, multiple myeloma, treatment> and <bortezomib, dexamethasone, cotreatment> (treatment is categorized in the Negative_Correlation). Other associations between two concepts are either implicit (e.g. variants frequently located within a gene) or rarely discussed. Accordingly, in this work we focus on annotating those eight concept pairs, as shown in solid lines in Figure 2A. To further characterize relations between entity pairs, we used eight biologically meaningful and nondirectional relation types (e.g. positive correlation; negative correlation) in our corpus as shown in Figure 2B. The details of

the relation types are described in our annotation guidelines.

Annotation process

To be consistent with previous annotation efforts, we randomly sampled articles from several existing datasets (i.e. NCBI Disease [19], NLM-Gene [14], GNormPlus [12], BC5CDR [9] and tmVar [20, 60]). A small set of PubMed articles were first used to develop our annotation guidelines and familiarize our annotators with both the task and TeamTat [61], a web-based annotation tool equipped to manage team annotation projects efficiently. Following previous practice in biomedical corpus development, we developed our annotation guidelines and selected PubMed articles consistently with previous studies. Furthermore, to accelerate entity annotation, we used previous annotations combined with automated preannotations (i.e. PubTator [28]), which can then be edited based

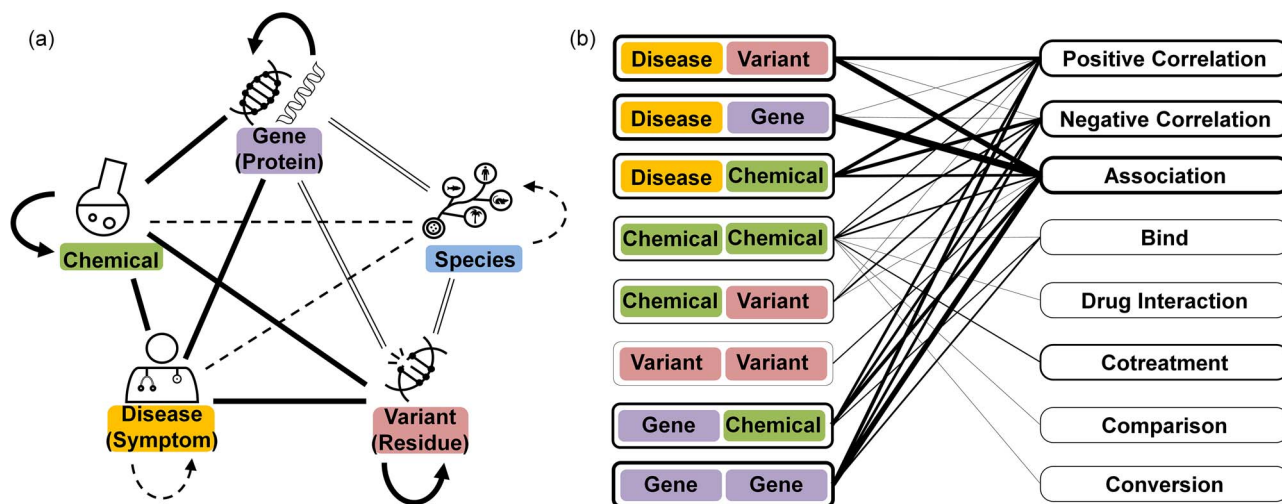


Figure 2. Relations annotated in BioRED corpus. (A) Categorized relations between concepts. The patterns of the lines between the concepts present the categories: (—) Popular associations: The concept pairs are frequently discussed in the biomedical literature. (---) Implied associations, e.g. the name of a gene can imply the corresponding species. (.....) Rarely discussed associations: Some other relation types are rarely discussed in the biomedical text (and this is why the concept of Cell Line is not listed here). (B) The mapping between the concept pairs and the relation types. The frame widths of the concept pairs/relation types and the bold lines between the two sides proportionally represent the frequencies.

on human judgment. Unlike entity annotation, each relation is annotated from scratch by hand with an appropriate relation type, except the chemical-induced-disease relations that were previously annotated in BC5CDR.

Every article in the corpus was first annotated by three annotators with a background in biomedical informatics to prevent erroneous and incomplete annotations (especially relations) due to manual annotation fatigue. If an entity or a relation cannot be agreed upon by the three annotators, this annotation was then reviewed by another senior annotator with a background in molecular biology. For each relation, two additional biologists assessed whether it is a novel finding versus background information and made the annotation accordingly. We annotated the entire set of 600 abstracts in 30 batches of 20 articles each. For each batch, it takes approximately 2 h per annotator to annotate entities, 8 h for relations and 6 h for assigning novel versus background label. The details of the data sampling and annotation rules are described in our annotation guidelines.

Data characteristics

The BioRED corpus contains a total of 20 419 entity mentions, corresponding to 3869 unique concept identifiers. We annotated 6503 relations in total. The proportion of novel relations among all annotated relations in the corpus is 69%. Table 4 shows the numbers of the entities (mentions and identifiers) and relations in the training, development and test sets.

In addition, we computed the inter-annotator-agreement (IAA) for the entity, relation and novelty annotations, where we achieved 97.01, 77.91 and 85.01%, respectively. Figure 3 depicts the distribution of the different concept pairs in the relations.

We also analyzed dataset statistics per document. The average document length consists of 11.9 sentences or

304 tokens. Furthermore, 34 entity spans (3.8 unique entity identifiers) and 10.8 relations are annotated per document. Among the relation types, 52% are associations, 27% are positive correlations, 17% are negative correlations and 2% are involved in the triple relations (e.g. two chemicals co-treat a disease).

Benchmarking methods

To assess the utility and challenges of the BioRED corpus, we conducted experiments to show the performance of leading RE models. For the NER task, each mention span was considered separately. We evaluate three state-of-the-art NER models on the corpus including bidirectional long short-term memory-conditional random field (BiLSTM-CRF), BioBERT-CRF and PubMedBERT-CRF. The input documents are first split into multiple sentences and encoded into a hidden state vector sequence by BiLSTM [62], BioBERT [49] and PubMedBERT [47], respectively. The models predicted the label corresponding to each of the input tokens in the sequence, then computed the network score using a fully connected layer and decode the best path of the tags in all possible paths by using CRF [63]. Here, we used the BIO (Begin, Inside, Outside) tagging scheme for the CRF layer.

We chose two BERT-based models, BERT-GT [64] and PubMedBERT [47], for evaluating the performance of current RE systems on the BioRED corpus. The first model is BERT-GT, which defines a graph transformer by integrating a neighbor-attention mechanism into the BERT architecture to avoid the effect of the noise from the longer text. BERT-GT was specifically designed for document-level relation extraction tasks and utilizes the entire sentence or passage to calculate the attention of the current token, which brings significant improvement to the original BERT model. PubMedBERT is a trained biomedical language model based on transformer architecture. It is

Table 4. Number of entity (mention and identifier) and relation annotations in the BioRED corpus, the IAA and the distribution between the training, development and test sets. The parenthesized numbers are the unique entities linked with concept identifiers.

Annotation s		Training	Dev	Tests	Total	IAA
Document		400	100	100	600	-
Entity (ID)	All	13 351 (2708)	3533 (956)	3535 (982)	20 419 (3869)	97.01%
	Gene	4430 (1141)	1087 (368)	1180 (399)	6697 (1643)	97.35%
	Disease	3646 (576)	982 (244)	917 (244)	5545 (778)	96.06%
	Chemical	2853 (486)	822 (184)	754 (170)	4429 (651)	96.12%
	Variant	890 (420)	250 (135)	241 (137)	1381 (678)	97.79%
	Species	1429 (37)	370 (13)	393 (11)	2192 (47)	99.43%
	Cell Line	103 (48)	22 (12)	50 (21)	175 (72)	99.68%
Relation		4178	1162	1163	6503	77.91%
Relation pair with novelty findings		2838	835	859	4532	85.01%

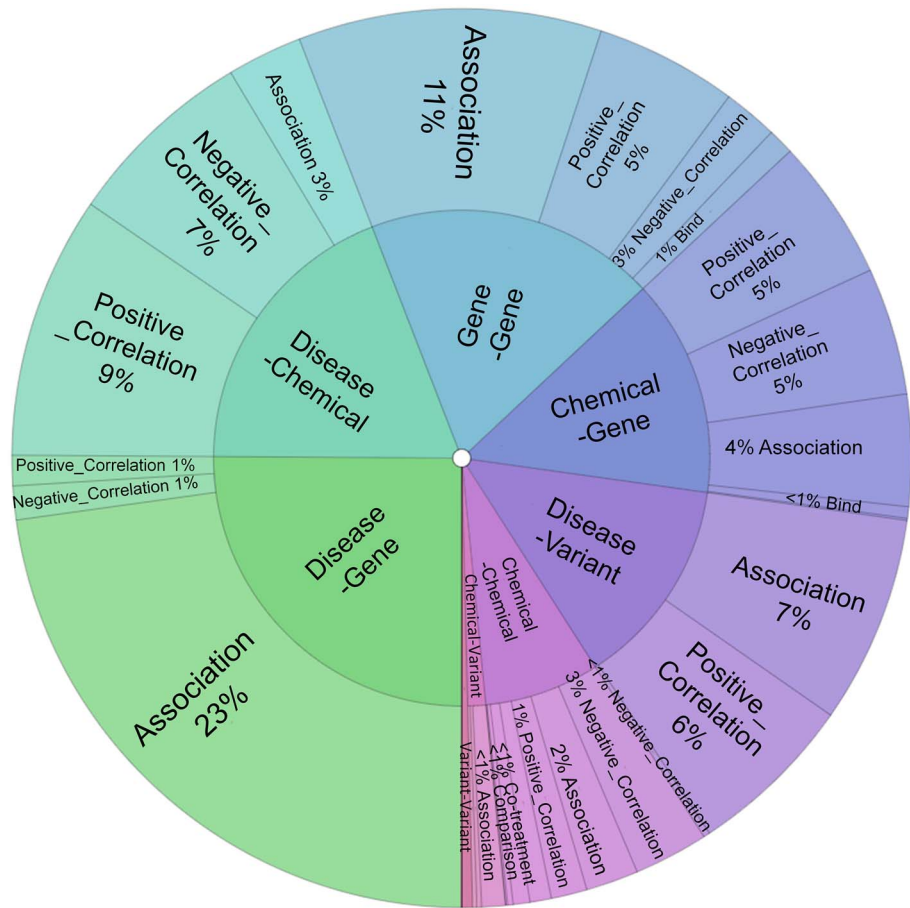


Figure 3. The distribution of concept pairs and relation types in the BioRED corpus.

currently a state-of-the-art text-mining method, which applies the biomedical domain knowledge (biomedical text and vocabulary) for the BERT pretrained language model. In the benchmarking, we used the text classification framework for the RE model development.

For both NER and RE evaluations, the training and development sets were first used for model development and parameter optimization before a trained model is evaluated on the test set. Benchmark implementation details are provided in Supplementary Materials A.1. Standard precision, recall and F-score metrics are used. To allow approximate entity matching, we also applied

relaxed versions of the F-score to evaluate NER. In this case, as long as the boundary of the predicted entity overlaps with the gold standard span, it is considered as a successful prediction.

Results

NER results on the test set

Table 5 shows the evaluation of NER on the test set. The first run is evaluated by strict metrics. The concept type and boundary of the entity should exactly match the entity in the text. The second run is evaluated by

Table 5. Performance of NER models on test set. All numbers are F-scores.

Metrics	Methods	All	Gene	Disease	Chemical	Species	CellLine	Variant
Strict	BiLSTM-CRF	87.1	87.3	83.3	88.2	96.3	80.9	82.9
	BioBERT-CRF	88.7	89.5	84.8	89.7	96.7	83.5	83.9
	PubMedBERT-CRF	89.3	92.4	83.5	88.6	97.0	90.5	87.3
Relaxed	BiLSTM-CRF	92.4	92.3	92.2	91.9	96.8	85.4	93.6
	BioBERT-CRF	93.4	93.8	93.6	91.3	97.0	90.1	92.3
	PubMedBERT-CRF	93.5	94.7	92.6	91.1	97.0	92.6	94.5

relaxed metrics: The entity boundary should overlap, and the same entity type is required. Unlike BiLSTM-CRF, the BERT-based methods contain well pretrained language models for extracting richer features, hence achieving better performance overall. Further, PubMedBERT performs even better than BioBERT on genes, variants and cell lines. BioBERT uses the original BERT model’s vocabulary generated from general domain text, which causes a lack of understanding of the biomedical entities. On the contrary, PubMedBERT generates the vocabulary from scratch using biomedical text, and it achieves the highest F-score (89.3% in strict metrics). Among these entity types, the PubMedBERT-CRF achieves the highest performance of 97% in F1 score to species entity recognition as less term ambiguity and variation issues are found in species names.

RE results on the test set

We also evaluated performance on the RE task by different benchmark schemas: (i) entity pair: to extract the pair of concept identifiers within the relation, (ii) entity pair + relation type: to recognize the specific relation type for the extracted pairs and (iii) entity pair + relation type + novelty: to further label the novelty for the extracted pairs. In this task, the gold-standard concepts in the articles are given. We applied BERT-GT and PubMedBERT to recognize the relations and the novelty in the test set.

As shown in Table 6, the overall performance of PubMedBERT is higher than that of BERT-GT in all schemas. Because the numbers of relations in <D,V>, <C,V> and <V,V> are low, their performance is not comparable to that of other concept pairs, especially <V,V> (the F-score is 0% for two models). In the first schema, BERT-GT and PubMedBERT can achieve performance above 72% for the F-scores, which is expected and promising in the document-level RE task. To predict the relation types (e.g. positive correlation) other than entity pairs, however, is still quite challenging. The best performance on the second schema is only 58.9%, as the number of instances in many relation types is insufficient. The performances of different relation types of our best model using PubMedBert are provided in Supplementary Materials A.2. The performance on the third schema dropped to 47.7%. In some cases, the statements of the relations in abstracts are usually concise, and the details of the related mechanism can only be found in the full text.

Benefits of multiple entity recognition and relation extraction

To test the hypothesis that our corpus can result in a single model with better performance, we trained multiple separate NER and RE models, each with an individual concept (e.g. gene) or relation (e.g. gene–gene) for comparison. We used PubMedBERT for this evaluation since it achieved the best performances in both the NER and RE tasks. As shown in Table 7, both models trained on all entities or relations generally perform better than the models trained on most of the entities or relations, while the improvement for RE is generally larger. The performance on NER and RE tasks is both obviously higher in the single model. Especially for entities and relations (e.g. cell lines and chemical–chemical relations) with insufficient amounts, the model trained on multiple concepts/relations can obtain larger improvements. The experiment demonstrated that training NER/RE models with more relevant concepts or relations not only can reduce resource usage but also can achieve better performance.

Discussion

The relaxed NER results in Table 5 for overall entity type are over 92% for all methods, suggesting the maturity of current tools for this task. If considering the performance of each concept individually, the recognition of genes, species and cell lines can reach higher performance (over 90% in strict F-score) since the names are often simpler and less ambiguous than other concepts. The best model for genomic variants achieves an F-score of 87.3% in strict metrics and 94.5% in relaxed metrics, which suggests that the majority of the errors are due to incorrect span boundaries. Most variants are not described in accordance with standard nomenclature (e.g. ‘ACG—> AAG substitution in codon 420’), thus it is difficult to exactly identify the boundaries. Similar to genomic variants, diseases are difficult to be identified due to term variability and most errors are caused by mismatched boundaries. For example, our method recognized a part (‘papilledema’) of a disease mentioned (‘bilateral papilledema’) in the text. Disease names also present greater diversity than other concepts: 55.4% of the disease names in the test set are not present in the training/development sets. Chemical names are extremely ambiguous with

Table 6. Performance on RE task for the first schema: extracting the entity pairs within a relation, second schema: extracting the entity pairs and the relation type and the third schema: further labeling the novelty for the extracted pairs. All numbers are F-scores. The <G,D> is the concept pair of the gene (G) and the disease (D). The columns of those entity pairs present the RE performance in F-scores.

Eval Schema	Methods	All	<G,D>	<G,G>	<G,C>	<D,V>	<C,D>	<C,V>	<C,C>
Entity pair	BERT-GT	72.1	63.8	78.5	77.7	69.8	76.2	58.8	74.9
	PubMedBERT	72.9	67.2	78.1	78.3	67.9	76.5	58.1	78.0
+Relation type	BERT-GT	56.5	54.8	63.5	60.2	42.5	67.0	11.8	52.9
	PubMedBERT	58.9	56.6	66.4	59.9	50.8	65.8	25.8	54.4
+Novelty	BERT-GT	44.5	37.5	47.3	55.0	36.9	51.9	11.8	48.5
	PubMedBERT	47.7	40.6	54.7	54.8	42.8	51.6	12.9	50.3

G = gene, D = disease, V = variant and C = chemical.

Table 7. The comparison of the models trained on all entities/relations to the models trained on individual entity/relation. The <G,D> is the relation of the gene (G) and the disease (D). All models are evaluated by strict metrics.

Entities /Relations	Types	All entities or relations			Single entity or relation		
		Precision	Recall	F-score	Precision	Recall	F-score
Entity	G	92.2	92.5	92.4	90.8	91.0	90.9
	D	80.7	86.5	83.5	83.2	85.7	84.4
	C	87.9	89.3	88.6	87.3	92.4	89.8
	V	88.8	85.9	87.3	84.7	87.1	85.9
	S	95.8	98.2	97.0	95.2	96.4	95.8
	CL	95.6	86.0	90.5	77.1	74.0	75.5
Relation	<G,D>	63.6	71.2	67.2	75.8	62.7	68.7
	<G,G>	81.5	75.0	78.1	57.3	80.0	66.8
	<G,C>	74.1	83.1	78.3	66.7	68.9	67.8
	<D,V>	71.2	64.9	67.9	76.5	51.5	61.5
	<C,D>	73.3	79.9	76.5	78.2	85.2	81.5
	<C,V>	60.0	56.3	58.1	53.3	50.0	51.6
	<C,C>	75.3	80.9	78.0	64.2	72.3	68.0

G = gene, D = disease, C = chemical, V = variant, S = species and CL = cell line.

other concepts: half of the errors for chemicals are incorrectly labeled as other concept types (e.g. gene), since some chemicals are interchangeable with other concepts, such as proteins and drugs. Moreover, we merged the annotations matched by the dictionary to the results of the PubMedBERT-CRF model. However, the performance of the dictionary method heavily depends on the difficulties of the term variation and ambiguity issues. Especially, there are many ambiguous terms in the dictionary, such as ‘B1’, ‘Beta’ and ‘98–4.9’ in Cellosaurus. Although the F1 score of the dictionary cannot compete with the machine learning method, merging the results from both methods can improve the recall of all the concepts (see details in Supplementary Materials A.3).

Experimental results in Table 6 show that the RE task remains challenging in biomedicine, especially for the new task of extracting novel findings. In our observation, there are three types of errors in novelty identification. First, some abstracts do not indicate which concept pairs represent novel findings, and instead, provide more details in the full text. Such cases confused both the human annotators and the computer algorithms. Second, when the mechanism of interaction between two relevant entities is unknown, and the study aims to investigate it but the hypothesized mechanism is shown to be false. Third, the authors

frequently mention relevant background knowledge within their conclusion. As an example, ‘We conclude that Rg1 may significantly improve the spatial learning capacity impaired by chronic morphine administration and restore the morphine-inhibited LTP. This effect is NMDA receptor-dependent.’ in the conclusion of the PMID:18308784, the Rg1 responded to morphine as background knowledge. But it is mentioned together with the novelty knowledge pair <Rg1, NMDA receptor>. In this case, our method misclassified the pair < Rg1, morphine> as novel. We also conducted an experiment to evaluate the effect of section information for novelty detection. The experimental results show that the structured section information (e.g. TITLE, PURPOSE, METHODS, RESULTS, ...) can be useful for novelty classification by boosting the best F1 score from 47.7% to 48.9% (see details in Supplementary Materials A.4). However, this result was obtained on a subset of 191 abstracts with structured section information due to limited availability.

The results in Table 7 demonstrate that training NER/RE models on one rich dataset with multiple concepts/relations simultaneously can not only make the trained model simpler and more efficient, but also more accurate. More importantly, we notice that for the entities and relations with a lower number of

training instances (e.g. cell lines and chemical–chemical relations), simultaneous prediction is especially beneficial for improving performance. Additionally, merging entity results from different models often poses some challenges, such as ambiguity or overlapping boundaries between different concepts.

Conclusion

In the past, biomedical RE datasets were typically built for a single entity type or relation. To enable the development of RE tools that can accurately recognize multiple concepts and their relations in biomedical texts, we have developed BioRED, a high-quality RE corpus, with one-of-a-kind novelty annotations. Similar to other commonly used biomedical datasets, e.g., BC5CDR [9], we expect BioRED to serve as a benchmark for not only biomedical-specific NLP tools but also for the development of RE methods in the general domain. Additionally, the novelty annotation in BioRED proposes a new NLP task that is critical for information extraction in practical applications. Recently, the dataset was successfully used by the NIH LitCoin NLP Challenge (<https://ncats.nih.gov/funding/challenges/litcoin>) and a total of over 200 teams participated in the Challenge.

This work has implications for several real-world use cases in medical information retrieval, data curation and knowledge discovery. Semantic search has been commonly practiced in the general domain but much less so in biomedicine. For instance, several existing studies retrieve articles based on the co-occurrence of two entities [65–68] or rank search results by co-occurrence frequency. Our work could accelerate the development of semantic search engines in medicine. Based on the extracted relations within documents, search engines can semantically identify articles by two entities with relations (e.g. 5-FU-induced cardiotoxicity) or by expanding the user queries from an entity (e.g. 5-FU) to the combination of the entity and other relevant entities (e.g. cardiotoxicity, diarrhea).

While BioRED is a novel and high-quality dataset, it has a few limitations. First, we are only able to include 600 abstracts in the BioRED corpus due to the prohibitive cost of manual annotation and limited resources. Nonetheless, our experiments show that except for a few concept pairs and relation types (e.g. variant–variant relations) that occur infrequently in the literature, its current size is appropriate for building RE models. Our experimental results in Table 7 also show that in some cases, the performance of entity class with a small number of training instances (e.g. Cell Line) can be significantly boosted when training together with other entities in one corpus. Second, the current corpus is developed on PubMed abstracts, as opposed to the full text. While full text contains more information, data access remains challenging in real-world settings. More investigation is warranted on this topic in the future.

Key Points

- First review on publicly available biomedical named entity recognition and relation extraction (RE) datasets.
- We present a first-of-its-kind biomedical relation extraction dataset (BioRED) with multiple entity types and relation pairs at the document level.
- The novelty RE task is proposed to differentiate between a novel finding or previously known background knowledge.
- Several cutting-edge deep learning models are evaluated on BioRED, and results show that there is much room for improvement for the RE task.

Supplementary data

Supplementary data are available online at <http://bib.oxfordjournals.org/>.

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References

1. Singhal A, Simmons M, Lu Z. Text mining genotype-phenotype relationships from biomedical literature for database curation and precision medicine. *PLoS Comput Biol* 2016;**12**(11):e1005017.
2. Lee K, Lee S, Park S, et al. BRONCO: Biomedical entity relation oncology corpus for extracting gene-variant-disease-drug relations. *Database* 2016;**2016**:baw043.
3. Wei C-H, Peng Y, Leaman R, et al. Assessing the state of the art in biomedical relation extraction: overview of the BioCreative V chemical-disease relation (CDR) task. *Database* 2016;**2016**:baw032.
4. Baptista D, Ferreira PG, Rocha M. Deep learning for drug response prediction in cancer. *Brief Bioinform* 2021;**22**(1):360–79.
5. Kim J-D, Ohta T, Tateisi Y, et al. GENIA corpus—a semantically annotated corpus for bio-textmining. *Bioinformatics* 2003;**19**(suppl_1):i180–2.
6. Pyysalo S, Ginter F, Heimonen J, et al. BioInfer: a corpus for information extraction in the biomedical domain. *BMC Bioinform* 2007;**8**(1):1–24.
7. Krallinger M, Leitner F, Rodriguez-Penagos C, et al. Overview of the protein-protein interaction annotation extraction task of BioCreative II. *Genome Biol* 2008;**9**(2):1–19.
8. Herrero-Zazo M, Segura-Bedmar I, Martínez P, et al. The DDI corpus: An annotated corpus with pharmacological substances and drug–drug interactions. *J Biomed Inform* 2013;**46**(5):914–20.

9. Li J, Sun Y, Johnson RJ, et al. BioCreative V CDR task corpus: a resource for chemical disease relation extraction. *Database* 2016;**2016**:baw068.
10. Krallinger M, Rabal O, Akhondi SA, et al. Overview of the BioCreative VI chemical-protein interaction Track. In: *Proceedings of the sixth BioCreative Challenge Evaluation Workshop*, Bethesda, MD USA: BioCreative, 2017, 142–147.
11. Wang X, Lyu J, Dong L, et al. Multitask learning for biomedical named entity recognition with cross-sharing structure. *BMC Bioinform* 2019;**20**(1):1–13.
12. Wei C-H, Kao H-Y, Lu Z. GNormPlus: an integrative approach for tagging genes, gene families, and protein domains. *Biomed Res Int* 2015;**2015**:918710.
13. Akdemir A, Shibuya T. Analyzing the effect of multi-task learning for biomedical named entity recognition. arXiv preprint arXiv:2011.00425. 2020.
14. Islamaj Doğan R, Wei C-H, Cissel D, et al. NLM-Gene, a richly annotated gold standard dataset for gene entities that addresses ambiguity and multi-species gene recognition. *J Biomed Inform* 2021;**118**:103779.
15. Morgan AA, Lu Z, Wang X, et al. Overview of BioCreative II gene normalization. *Genome Biol* 2008;**9**(2):1–19.
16. Hirschman L, Colosimo M, Morgan A, et al. Overview of BioCreAtIvE task 1B: normalized gene lists. *BMC Bioinform* 2005;**6**(1):S11.
17. Islamaj Doğan R, Leaman R, Kim S, et al. NLM-Chem, a new resource for chemical entity recognition in PubMed full text literature. *Sci Data* 2021;**8**(1):1–12.
18. Krallinger M, Rabal O, Leitner F, et al. The ChEMDNER corpus of chemicals and drugs and its annotation principles. *J Chem* 2015;**7**(1):1–17.
19. Islamaj Doğan R, Leaman R, Lu Z. NCBI disease corpus: a resource for disease name recognition and concept normalization. *J Biomed Inform* 2014;**47**:1–10.
20. Wei C-H, Harris BR, Kao H-Y, et al. tmVar: a text mining approach for extracting sequence variants in biomedical literature. *Bioinformatics* 2013;**29**(11):1433–9.
21. Doughty E, Kertesz-Farkas A, Bodenreider O, et al. Toward an automatic method for extracting cancer-and other disease-related point mutations from the biomedical literature. *Bioinformatics* 2011;**27**(3):408–15.
22. Caporaso JG, Baumgartner WA, Jr, Randolph DA, et al. Mutation-Finder: a high-performance system for extracting point mutation mentions from text. *Bioinformatics* 2007;**23**(14):1862–5.
23. Pafilis E, Frankild SP, Fanini L, et al. The SPECIES and ORGANISMS resources for fast and accurate identification of taxonomic names in text. *PLoS One* 2013;**8**(6):e65390.
24. Gerner M, Nenadic G, Bergman CM. LINNAEUS: a species name identification system for biomedical literature. *BMC Bioinform* 2010;**11**(1):1–17.
25. Arighi C, Hirschman L, Lemberger T, et al. Bio-ID track overview. In: *BioCreative VI Challenge Evaluation Workshop*, Bethesda, MD USA: BioCreative, 2017, 14–19.
26. Kim J-D, Ohta T, Tsuruoka Y, et al. Introduction to the bio-entity recognition task at JNLPBA. In: *Proceedings of the International Joint Workshop on Natural Language Processing in Biomedicine and Its Applications*. Geneva, Switzerland: COLING, 2004, 70–75.
27. Bada M, Eckert M, Evans D, et al. Concept annotation in the CRAFT corpus. *BMC Bioinform* 2012;**13**(1):1–20.
28. Wei C-H, Allot A, Leaman R, et al. PubTator central: automated concept annotation for biomedical full text articles. *Nucleic Acids Res* 2019;**47**(W1):W587–93.
29. Wei C-H, Lee K, Leaman R, et al. Biomedical mention disambiguation using a deep learning approach. In: *Proceedings of the 10th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, New York, United States: Association for Computing Machinery, 2019, 307–313.
30. Leaman R, Lu Z. TaggerOne: joint named entity recognition and normalization with semi-Markov Models. *Bioinformatics* 2016;**32**(18):2839–46.
31. Hendrickx I, Kim SN, Kozareva Z, et al. Semeval-2010 task 8: Multi-way classification of semantic relations between pairs of nominals. In: *Proceedings of the 5th International Workshop on Semantic Evaluation*, ACL 2010, Uppsala, Sweden: Association for Computational Linguistics, 2010, 33–38.
32. Zhang Y, Zhong V, Chen D, et al. Position-aware attention and supervised data improve slot filling. In: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, Copenhagen, Denmark: Association for Computational Linguistics, 2017, 35–45.
33. Walker C, Strassel S, Medero J, et al. ACE 2005 multilingual training corpus. In: *Linguistic Data Consortium*, Philadelphia: Linguistic Data Consortium, 2006.
34. Yao Y, Ye D, Li P, et al. DocRED: a large-scale document-level relation extraction dataset. In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Florence, Italy: Association for Computational Linguistics, 2019, 764–777.
35. Dong K, Zhao Y, Sun A, et al. DocOIE: a document-level context-aware dataset for OpenIE. In: *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, Association for Computational Linguistics, 2021, 2377–2389.
36. Ding J, Berleant D, Nettleton D, et al. Mining MEDLINE: abstracts, sentences, or phrases? In: *Biocomputing 2002*. Kauai, Hawaii, USA: World Scientific, 2001, 326–37.
37. Bunesco R, Ge R, Kate RJ, et al. Comparative experiments on learning information extractors for proteins and their interactions. *Artif Intell Med* 2005;**33**(2):139–55.
38. Nédellec C. Learning language in logic-genic interaction extraction challenge. In: *4. Learning Language in Logic Workshop (LLL05)*. Born, Germany: ACM-Association for Computing Machinery, 2005.
39. Fundel K, Küffner R, Zimmer R. RelEx—relation extraction using dependency parse trees. *Bioinformatics* 2007;**23**(3):365–71.
40. Miranda A, Mehryary F, Luoma J, et al. Overview of DrugProt BioCreative VII track: quality evaluation and large scale text mining of drug-gene/protein relations. In: *Proceedings of the BioCreative VII Challenge Evaluation Workshop*, BioCreative, 2021, 11–21.
41. Airola A, Pyysalo S, Björne J, et al. All-paths graph kernel for protein-protein interaction extraction with evaluation of cross-corpus learning. *BMC Bioinform* 2008;**9**(11):1–12.
42. Peng Y, Rios A, Kavuluru R, et al. Extracting chemical–protein relations with ensembles of SVM and deep learning models. *Database* 2018;**2018**:bay073.
43. Yadav S, Ekbal A, Saha S, et al. Feature assisted stacked attentive shortest dependency path based Bi-LSTM model for protein–protein interaction. *Knowledge-Base Syst* 2019;**166**:18–29.
44. Luo L, Yang Z, Cao M, et al. A neural network-based joint learning approach for biomedical entity and relation extraction from biomedical literature. *J Biomed Inform* 2020;**103**:103384.
45. Li Y, Chen Y, Qin Y, et al. Protein-protein interaction relation extraction based on multigranularity semantic fusion. *J Biomed Inform* 2021;**123**:103931.
46. Raj Kanakarajan K, Kundumani B, Sankarasubbu M. BioELECTRA: pretrained biomedical text encoder using discriminators.

- In: *Proceedings of the 20th Workshop on Biomedical Language Processing*, Association for Computational Linguistics, 2021, 143–154.
47. Gu Y, Tinn R, Cheng H, et al. Domain-specific language model pretraining for biomedical natural language processing. *ACM Trans Comput Healthc* 2021;**3**(1):1–23.
 48. Alrowili S, Vijay-Shanker K. BioM-transformers: building large biomedical language models with BERT, ALBERT and ELECTRA. In: *Proceedings of the 20th Workshop on Biomedical Language Processing*. Association for Computational Linguistics, 2021, 221–227.
 49. Lee J, Yoon W, Kim S, et al. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 2020;**36**(4):1234–40.
 50. Xenarios I, Fernandez E, Salwinski L, et al. DIP: the database of interacting proteins: 2001 update. *Nucleic Acids Res* 2001;**29**(1): 239–41.
 51. Gurulingappa H, Rajput AM, Roberts A, et al. Development of a benchmark corpus to support the automatic extraction of drug-related adverse effects from medical case reports. *Journal of Biomedical Informatics* 2012;**45**(5):885–92.
 52. Henry S, Buchan K, Filannino M, et al. 2018 n2c2 shared task on adverse drug events and medication extraction in electronic health records. *J Am Med Inform Assoc* 2019;**27**(1):3–12.
 53. Aronson AR. *Effective mapping of biomedical text to the UMLS Metathesaurus: the MetaMap program*. In: *Proceedings of the AMIA Symposium*. Washington, DC, USA: American Medical Informatics Association, 2001, 17–21.
 54. Su J, Wu Y, Ting H-F, et al. RENET2: high-performance full-text gene-disease relation extraction with iterative training data expansion. *NAR Genomics. Bioinformatics* 2021;**3**(3):lqab062.
 55. Wu Y, Luo R, Leung H, et al. Renet: A deep learning approach for extracting gene-disease associations from literature. In: *International Conference on Research in Computational Molecular Biology*. Washington, DC, USA: Springer, 2019, 272–284.
 56. Peng N, Poon H, Quirk C, et al. Cross-sentence n-ary relation extraction with graph lstms. *Trans Assoc Comput Linguist* 2017;**5**: 101–15.
 57. Kim J-D, Ohta T, Pyysalo S, et al. Overview of BioNLP'09 shared task on event extraction. In: *Proceedings of the BioNLP 2009 Workshop Companion Volume for Shared Task*, Boulder, Colorado: Association for Computational Linguistics, 2009, 1–9.
 58. Kim J-D, Wang Y, Takagi T, et al. Overview of genia event task in bionlp shared task 2011. In: *Proceedings of BioNLP shared task 2011 workshop*, Portland, Oregon, USA: Association for Computational Linguistics, 2011, 7–15.
 59. Pyysalo S, Ohta T, Rak R, et al. Overview of the cancer genetics and pathway curation tasks of bionlp shared task 2013. *BMC Bioinform* 2015;**16**(10):1–19.
 60. Wei C-H, Phan L, Feltz J, et al. tmVar 2.0: integrating genomic variant information from literature with dbSNP and ClinVar for precision medicine. *Bioinformatics* 2018;**34**(1):80–7.
 61. Islamaj Doğan R, Kwon D, Kim S, et al. TeamTat: a collaborative text annotation tool. *Nucleic Acids Res* 2020;**48**(W1): W5–11.
 62. Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput* 1997;**9**(8):1735–80.
 63. Lafferty J, McCallum A, Pereira FC. *Conditional random fields: probabilistic models for segmenting and labeling sequence data*. In: *Proceedings of the Eighteenth International Conference on Machine Learning*. San Francisco, CA, United States: Association for Computing Machinery, 2001, 282–289.
 64. Lai P-T, Lu Z. BERT-GT: cross-sentence n-ary relation extraction with BERT and Graph Transformer. *Bioinformatics* 2020;**36**(24): 5678–85.
 65. Allot A, Peng Y, Wei C-H, et al. LitVar: a semantic search engine for linking genomic variant data in PubMed and PMC. *Nucleic Acids Res* 2018;**46**(W1):W530–6.
 66. Thomas P, Starlinger J, Vowinkel A, et al. GeneView: a comprehensive semantic search engine for PubMed. *Nucleic Acids Res* 2012;**40**(W1):W585–91.
 67. Dörpinghaus J, Klein J, Darms J, et al. SCAIView-A Semantic Search Engine for Biomedical Research Utilizing a Microservice Architecture. In: *SEMANTICS Posters&Demos*, Vienna, Austria: CEUR-WS, 2018.
 68. Pang X, Bou-Dargham MJ, Liu Y, et al. Accelerating cancer research using big data with BioKDE platform. In: *Proceedings of the American Association for Cancer Research Annual Meeting*, Chicago, IL. Philadelphia (PA): AACR, 2018.